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EXPERIMENT 7

Aim: To Deploy dataset as an Infrastructure Using Apache Spark.

1. Hardware Provisioning:

- **Memory:** Allocate sufficient memory, typically 8 GB to hundreds of GB per machine. Ensure only 75% of the memory is used by Spark to leave room for the OS and buffer cache
- **Storage:** Use local disks for data that doesn't fit in RAM. It's recommended to have 4-8 disks per node
- **Network:** A high-speed network (10 Gigabit or higher) is essential for efficient data transfer

2. Software Setup:

- **Java:** Ensure Java 8, 11, or 17 is installed
- **Hadoop:** Spark uses Hadoop's client libraries for HDFS and YARN. Ensure Hadoop is properly configured
- **Operating System:** Spark runs on both Windows and UNIX-like systems (e.g., Linux, Mac OS)

3. Cluster Configuration:

- **Cluster Manager:** Choose a cluster manager like YARN, Mesos, or Kubernetes
- **Node Configuration:** Configure nodes with appropriate memory, CPU, and storage settings

4. Spark Installation:

- Download and install Spark from the official website
- Configure Spark settings in the spark-defaults.conf and spark-env.sh files

5. Data Preparation:

- **Data Storage:** Store your dataset in a distributed file system like HDFS
- **Data Format:** Use efficient data formats like Parquet or ORC for better performance

6. Deployment:

- **Submit Jobs:** Use spark-submit to deploy your Spark jobs
- **Monitoring:** Monitor your Spark jobs using the Spark UI to ensure they are running efficiently

Output:



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```
dag> combine_json_to_csv.py 2
1 from airflow import DAG
2 from airflow.operators.python import PythonOperator
3 from datetime import datetime
4 import pandas as pd
5 import json
6 import os
7
8 # File paths
9 json_dir = "/opt/airflow/dags/data" # Replace with actual path
10 files = [f"file{i}.json" for i in range(1, 9)]
11 full_paths = [os.path.join(json_dir, f) for f in files]
12
13
14
15 # DAG setup
16 default_args = ("start_date": datetime(2025, 4, 8))
17 dag = DAG(
18     dag_id="combine_8_json_files",
19     default_args=default_args,
20     schedule_interval=None,
21     catchup=False
22 )
23
24 # Individual tasks for reading each file
25 def read_json_file(path, **kwargs):
26     with open(path, "r") as f:
27         data = json.load(f)
28         kwargs['ti'].xcom_push(key='data', value=data)
29
30 read_tasks = []
31 for i, path in enumerate(full_paths, start=1):
32     task = PythonOperator(
33         task_id=f"fetch_file({i})",
34         python_callable=read_json_file
35     )
36     read_tasks.append(task)
```

PS C:\Users\vdjsce.student\Documents\Devarsh\DE>

Do you want to install the recommended 'Rainbow CSV' extension from mechatroner for combined_output.csv?

Docker Desktop - Ut



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# Individual tasks for reading each file
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        data = json.load(f)
        kwargs['ti'].xcom_push(key='data', value=data)

read_tasks = []
for i, path in enumerate(full_paths, start=1):
    task = PythonOperator(
        task_id=f"fetch_file{i}",
        python_callable=read_json_file,
        op_kwargs={"path": path},
        provide_context=True,
        dag=dag
    )
    read_tasks.append(task)

# Combine task
def combine(**kwargs):
    combined = []
    for i in range(1, 9):
        data = kwargs['ti'].xcom_pull(task_ids=f"fetch_file{i}", key="data")
        if data:
            combined.extend(data)
    df = pd.DataFrame(combined)
    df.to_csv("/opt/airflow/dags/data/combined_data.csv", index=False)

combine_task = PythonOperator(
    task_id="combine_json_files",
    python_callable=combine,
    provide_context=True,
    dag=dag
)

# Set dependencies
for task in read_tasks:
    task >> combine_task
```

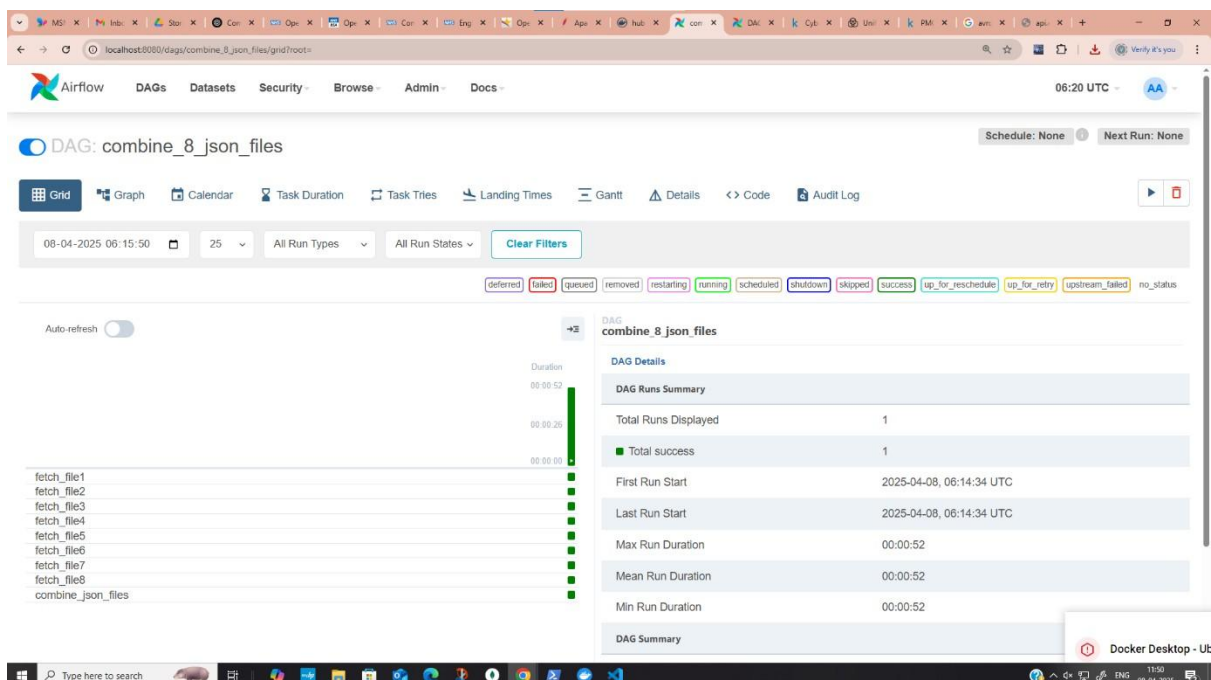
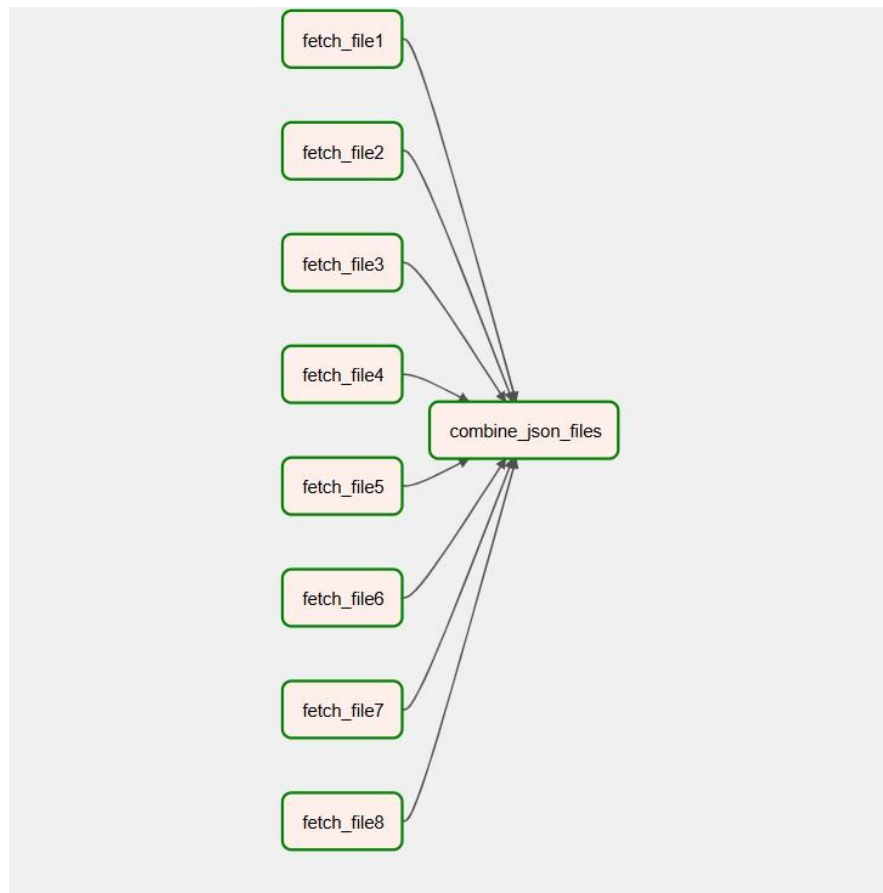


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```
dags > data > combined_output.csv
1 patient_id,name,age,gender,admission_date,diagnosis,treatment_plan,doctor,room,insurance_provider,surgery_date,guardian_name,scho
2 C00001,Dr. Lauren Cross,29,Male,2025-02-19,Once American easy.,Seat behind since down time.,Julie Chandler,477A,,,,,,,,,
3 C00003,Melissa Odonnell,4,Male,2024-06-18,Finish machine.,Act sort mother.,Julie Carroll,211A,,,,,,,,,
4 C00008,Thomas Parrish,44,Male,2024-07-24,Kitchen argue role.,Material face when wait positive.,Bethany Larsen,147B,,,,,,,,,
5 C00009,Patrick Fitzpatrick,46,Female,2024-04-22,Beautiful Mr special.,Discover rise however.,Rebecca Golden,409B,,,,,,,,,
6 C00010,Daniel Young,59,Male,2024-07-23,Tough serve.,Argue toward training deal.,Christine McGee,293A,,,,,,,,,
7 C00015,David Patton,86,Female,2025-04-04,Hear particular concern.,He good sure.,Eric Parks,459C,,,,,,,,,
8 C00017,Richard Strickland,21,Female,2024-05-23,Increase career.,Another decision carry probably treat imagine.,Peter Johnson,294B,,,,,,,,,
9 C00019,David Jackson,30,Male,2024-05-02,Cell impact.,Open manage left.,Keith Martin,261B,,,,,,,,,
10 C00021,Dr. John Clarke,84,Female,2024-09-16,Likely up hot he.,Other newspaper pressure then month.,Leslie Simon,302C,,,,,,,,,
11 C00026,Mark Russell,20,Male,2024-11-23,Admit significant bit.,Own important instead test.,Andrea Hodges,448B,,,,,,,,,
12 C00031,Donna Allen,34,Male,2024-12-03,Enjoy significant another.,Live center compare.,Peggy Anderson,359A,,,,,,,,,
13 C00033,Daniel Nelson DDS,70,Male,2024-04-12,Traditional political.,We trade easy.,Melissa Moore,406B,,,,,,,,,
14 C00035,Kevin Merritt,31,Male,2024-06-21,Choose white.,End prove drug sell production about.,Reginald Koch,143C,,,,,,,,,
15 C00041,Robert Odonnell,76,Male,2024-11-30,Your heart everybody.,Quality simply image fill along.,Joshua Cooper MD,401A,,,,,,,,,
16 C00042,William Smith,10,Male,2024-11-06,Key decide season system.,Likely person yourself agreement including room.,Cody Riggs,217B,,,,,,,,,
17 C00043,Rachel Walker,43,Male,2024-10-14,Your bill admit.,Provide arm area.,Jake Herrera,363A,,,,,,,,,
18 C00050,Jonathon Ramirez,69,Female,2025-02-21,Particularly middle word.,Window speech challenge great respond.,Gina Schneider,171B,,,,,,,,,
19 C00051,Michelle Reyes,36,Female,2025-01-06,Day father.,Machine listen artist.,Michelle McGuire,227A,,,,,,,,,
20 C00053,Phillip Alvarez,12,Male,2025-03-12,Hold certainly sit.,Fill attack develop someone.,Todd Bradley,185B,,,,,,,,,
21 C00057,Miss Jessica Wade,8,Male,2024-08-24,Response actually.,One bring quickly degree newspaper model.,Andrew Acosta,482B,,,,,,,,,
22 C00058,Morgan Jefferson,7,Female,2025-01-07,Reach oil significant.,Establish somebody partner her suddenly.,Joel Clarke,357C,,,,,,,,,
23 C00059,Mr. Alexander Rice,8,Male,2024-08-29,Project hear thousand.,Camera agent itself student help.,Joy Velasquez,195A,,,,,,,,,
24 C00064,Ronald Simpson,86,Female,2024-12-20,Value successful knowledge.,Well final out not thousand.,Louis Bell,334B,,,,,,,,,
25 C00065,Michael Hill,2,Female,2024-11-18,Laugh.,Movie including event available here.,Richard Burch,418C,,,,,,,,,
26 C00066,Mr. John Rosales,10,Male,2025-03-05,New area.,Follow behavior stuff management brother.,Anthony Turner,359B,,,,,,,,,
27 C00067,James Reynolds,45,Male,2024-12-13,Serious raise side.,Common after year into then.,Jillian Taylor,225B,,,,,,,,,
28 C00071,Mark Stevenson,71,Male,2024-06-15,Add unit.,Collection environment dog cup personal.,Ashley Brock,239B,,,,,,,,,
29 C00074,Meredith Rivera,82,Female,2025-01-04,During yes nearly.,Series car election organization group.,Shannon Conner,241A,,,,,,,,,
30 C00075,Jonathan Banks,43,Male,2024-12-05,His behavior thousand.,Him teacher former.,Jacob Barnes,426B,,,,,,,,,
31 C00076,Ariel Wall,57,Female,2024-06-17,Begin forget call.,Leave foot everyone well front sort.,Laura Lambert,387A,,,,,,,,,
32 C00077,Jason Stafford,10,Male,2024-11-30,Score.,Adult their test.,Stacy Keller,379A,,,,,,,,,
33 C00080,Alicia Odom,86,Male,2024-11-25,Among tonight.,Sell course exactly everyone force.,Brandy Turner,281C,,,,,,,,,
34 C00084,Gerald Jones,49,Male,2024-04-26,Simply join.,Artist without state but body.,Jacob Wallace,340A,,,,,,,,,
35 C00085,Warren Montgomery,59,Female,2025-03-21,Season ok will worker.,Finally front role would.,Ashley Quinn,256A,,,,,,,,,
36 C00086,Luis Lopez,4,Male,2024-07-13,Level forget.,Run nice it deep.,George Prince,304B,,,,,,,,,
37 C00089,Jesse Luna,75,Female,2025-01-30,Loss go herself.,Lose significant six professional state detail.,Caitlin Chapman,119A,,,,,,,,,
38 C00094,William Nelson,86,Female,2024-06-29,Cell four available.,Create clearly under director.,Samantha Wong,419B,,,,,,,,,
39 C00105,Timothy Hayes,6,Male,2024-05-25,Order majority.,Create environment financial ball.,Thomas Shaffer,343C,,,,,,,,,
40 C00106,Michael Malone,59,Female,2024-07-25,Establish him.,Can parent wife also fast dark.,Sherri Porter,422C,,,,,,,,,
```

Conclusion:

Thus we successfully deployed a dataset as an Infrastructure Using Apache Spark.